WHAT SOFTWARE-WORK TELEMETRY IDENTIFIES: ATTAINABLE LABOUR BOUNDS AND DESIGN-BASED ESTIMATION

Abstract. Software-work telemetry records events whose relationship to human labour can change under AI assistance. We determine which labour totals are compatible with specified observations, keeping elapsed time to acceptance and retained output as separate endpoints. Exact interval totals and a per-person capacity constraint yield attainable labour bounds at a requested cutoff. For overlapping observations and shared capacity, partitioning time at observation endpoints and target cutoffs reduces the continuous allocation problem exactly to a finite linear programme. Its extrema delimit all attainable totals under the declared observations. Adding separate interval bounds can give a wider range. When labour is probability-sampled, fixed telemetry predictions can assist estimation through a design-unbiased residual correction. Its variance depends on prediction residuals, so inaccurate predictions can reduce precision. A marked-history representation assigns deadline acceptance, delayed retention assessment and fixed-horizon labour their respective observation times without assuming Markov dynamics or independent attempts. We prove these results analytically and check finite examples computationally. They inform a proposed cluster-randomised measurement pilot. No organisational data are analysed, and empirical calibration remains open. The new results are outside the retained historical Lean proof coverage.

1. What the observations determine

Suppose a team records three person-hours of work during a four-hour interval. How much occurred during its first two hours? If one person supplied the work, the answer lies between one and two person-hours: at most two fit after the cutoff, and at most two fit before it. Both limits are attainable. The interval total therefore identifies a range without identifying the allocation within it. Commit times alone may supply even less information, because the same event sequence can arise from different amounts of human activity.

The engineering question is how much labour the observations determine, and which additional measurements would narrow the answer. We address it for AI-assisted software work using all-role person-hours, elapsed time to first acceptance and reference output retained at a stated follow-up. These definitions account for unsuccessful work, concurrent activity and post-acceptance repair. The principal result represents all feasible labour totals by a finite linear programme whose constraints encode exact interval observations and shared per-person capacity. Minimising and maximising the desired total gives its identified range: every value between the extrema is attainable by an admissible allocation. Narrower bounds therefore require additional observations or justified restrictions on those allocations.

When exact allocation totals are unavailable, probability sampling supports design-unbiased estimates through known inclusion probabilities. Fixed telemetry predictions can improve precision if their residuals reduce variance under the sampling design. This creates a useful role for telemetry even when its events leave labour unidentified: independent allocation observations can correct prediction error. Deterministic identification intervals, sampling uncertainty and uncertainty in a causal intervention comparison consequently answer different questions.

Wright and Ziegler's Standard Coder approach [17] motivates the observation problem. It infers coding activity from commit timing and predicts the resulting labels from code-change features. Standard Coding Hours (SCH) can serve as a common reference index, but

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calibration to independently observed labour requires a separate argument. AI may alter both the event-to-activity relationship and the work represented by a code change. The SPACE framework places activity and efficiency within a broader account of developer productivity [8]. The CUPS taxonomy examines interaction time costs in AI-assisted programming [13]. Our narrower contribution is the labour identified or estimable under an explicit observation scheme, together with compatible timing and retained-output endpoints.

Existing AI studies address different parts of this measurement problem. Peng et al. examine completion time among completers of a programming exercise [14]. Becker et al. study experienced maintainers in familiar repositories with a forecast-adjusted log-time contrast [2]. Cui et al. study weekly completed pull requests in organisational experiments [6], with assignment used as an instrument for adoption in their June 2025 manuscript [5, Section 4.1 and Table 3]. These differences in populations, interventions and outcomes prevent treating their estimates as interchangeable parameters. METR's subsequent account of selection and concurrent-agent measurement difficulties further motivates separate labour and elapsed-time records [3]. We use these studies to locate the measurement questions. They supply no data for the calculations here.

Sections 2 and 3 define the observation problem and endpoints. Section 4 derives the labour bounds, and Section 5 connects sampled labour to telemetry-assisted estimation. Section 6 gives a prospective protocol for assessing the measurements before estimating an intervention effect. The finite reduction and attainability arguments specialise standard optimisation and sampling results to the stated allocation constraints. The appendices examine illustrative execution mechanisms, correct selected Standard Coder arguments and compare two implementations of repeated discrete accounting queries. The runtime comparison leaves the cost of solving the labour-bounds programme and team productivity unmeasured.

2. Why event telemetry alone need not identify labour

Let a minute-indexed hidden variable $Z_t \in \{0,1\}$ indicate an activity state, and let $C_t \in \{0,1\}$ record a commit event. A fitted event model specifies both activity dynamics and the mechanism that produces commits from activity. Interpreting $Z_t = 1$ as "coding" requires validation against independently measured labour. Even before that validation, distinct activity laws can produce the same event law.

Proposition 2.1 (Failure of unrestricted identification)**.** *Suppose the states Z_t are independent Bernoulli variables with parameter $\pi \in (0,1)$, and, independently across minutes and of the states, B_t are Bernoulli variables with parameter $c \in (0,1)$. Define $C_t = Z_t B_t$. The complete law of the event process depends on (π, c) only through πc. In particular, $(\pi, c) = (1/2, 1/5)$ and $(1/4, 2/5)$ give identical observed laws but different expected active times.*

Proof. Each C_t is Bernoulli with parameter $\theta = \pi c$, and the C_t are independent. For every $n \geq 1$ and every binary string $(e_1, \ldots, e_n)$,

$$\mathbb{P}(C_1 = e_1, \ldots, C_n = e_n) = \theta^{\sum_t e_t}(1-\theta)^{n-\sum_t e_t}. \tag{1}$$

Both parameter pairs give $\theta = 1/10$. In contrast,

$$\mathbb{E}\left[\sum_{t=1}^{n} Z_t\right] = n\pi, \tag{2}$$

which is $n/2$ minutes in the first model and $n/4$ in the second. Thus the same observable distribution corresponds to two different target values. □

This construction belongs to the independent-state subfamily of a two-state hidden Markov model: both transition rows give the same next-state probability. It shows that the unrestricted family is not globally identifiable. Persistent or otherwise restricted models

may still be identifiable. Additional restrictions, informative code marks or independent activity observations may distinguish the relevant cases. A guarantee must state which of those restrictions or observations supplies the information. AI makes the question more immediate because commit production can continue while human attention is elsewhere.

We next ask how additional interval observations constrain labour. Appendix C shows why agreement between two model predictions cannot supply the required calibration.

3. Task accounting with explicit boundaries

3.1. **The task, the assigned intervention and the person-hour.** Enrol task i at time s_i, when its requested behaviour, eligibility and acceptance criteria are recorded. Give the task an identity tied to its requirement, so commits and pull requests can change without changing the unit counted. Before observing outcomes, define how split tasks, merged tasks and scope changes will be handled. A parent requirement can retain its original identity while children record implementation activity. Counting both the parent and its children as delivered output would count the same work twice. Changes affecting the stated requirement or acceptance criteria require a versioned record and a stated analysis rule.

That rule must also fix task granularity, acceptance credit and the follow-up clock before interventions are compared. Even additive reference scores can yield different deadline output when separately accepted components are regrouped into a parent accepted only after its last component. Freezing the score alone therefore leaves part of the comparison unspecified.

An *AI intervention* specifies available tools and model versions, eligible tasks, permitted autonomous actions, review responsibilities and release requirements. The proposed experiment assigns this complete specification. Record actual tool use separately, since difficult tasks may elicit more use and people may decline an assigned tool.

Let $a_{ijr}(t)$ be the fraction of person j's time allocated to task i in role r at time t. For a finite set of observed people, measurable allocation functions satisfy

$$a_{ijr}(t) \geq 0, \qquad \sum_{i,r} a_{ijr}(t) \leq 1, \qquad H_i(u) = \sum_{j,r} \int_{s_i}^{s_i+u} a_{ijr}(t)\, dt. \tag{3}$$

With time in hours, $H_i(u)$ has units of person-hours. Roles include specification, implementation, prompting, review, testing, integration and repair. Pair work can contribute two person-hours in one elapsed hour. For one person, two concurrent sessions cannot each receive a full hour of attention. State whether attended waiting belongs in the labour total. Unattended execution is recorded as agent resource use.

Equation (3) defines the target, while the observation procedure determines how precisely its allocations are known. Shared review and platform work therefore need an attribution rule, with an overhead category where the observations cannot justify allocation to particular tasks. Record agent instance-hours by resource class, tokens and charges separately from person-hours. Combining these resources requires an explicit economic conversion and changes the question from labour to cost.

Let T_i be elapsed time from enrolment to first acceptance. Starting this clock at enrolment includes queueing and any reasoning and specification between enrolment and the first commit. If the task is permanently abandoned before acceptance, put $T_i = \infty$. Abandonment is an outcome, whereas loss of observation is censoring. Record deployment separately whenever acceptance and deployment differ.

3.2. **A frozen reference and a follow-up assessment.** Assign a nonnegative reference score S_i^0 using a declared version and representation. It may count tasks, independent functional units or frozen SCH weights. A labour-calibrated reference would target

$$S^0(x) = \mathbb{E}[H^{\mathrm{ref}} \mid X = x] \tag{4}$$

for a specified reference population, resource boundary, quality criterion and stopping rule, provided the expectation is finite. A predictor trained only on imputed coding labels targets those labels. Its calibration to independently measured labour requires a separate empirical test. The superscript 0 denotes the fixed reference against which work under the intervention is assessed.

For a fixed-scope causal comparison, requirements recorded at intake are preferable to final code differences: AI can change the latter. A retrospective code score can still be reported, with its dependence on the implementation made explicit. A nonlinear scoring rule need not be additive, so splitting or squashing a change can alter its score. Score a canonical task representation once, or use an additive requirement rubric. A failed attempt, retry or revert must not generate another unit of retained output.

Choose an acceptance horizon $\tau > 0$ and a follow-up duration $\Delta \geq 0$. For accepted tasks, let $R_i(\Delta) \in [0, 1]$ be the assessed fraction of the accepted requirement still meeting the rubric at time $s_i + T_i + \Delta$. Retained reference output is

$$Q_i(\tau, \Delta) = \begin{cases} S_i^0 R_i(\Delta), & T_i \leq \tau, \\ 0, & T_i > \tau. \end{cases} \tag{5}$$

The rubric must distinguish an intentional replacement from a defect-driven loss and explain fractional scores. Surviving lines, successful tests and absence of recorded incidents are observations of quality. None establishes complete correctness. Reliability of the assessment itself therefore belongs in the study.

For a finite intake cohort with positive total labour, define

$$P_H(\tau, \Delta) = \frac{\sum_i Q_i(\tau, \Delta)}{\sum_i H_i(\tau + \Delta)}. \tag{6}$$

The denominator is the cohort's resource total through the fixed enrolment horizon. It includes labour on rejected, unfinished and abandoned tasks, together with task-attributable work after acceptance. An early acceptance leaves more of this horizon remaining than a late one. Stopping each task's repair accounting exactly Δ after acceptance therefore defines a different labour boundary. Either convention can be useful, but the choice must be explicit. Follow every enrolment for $\tau + \Delta$ to ascertain the numerator and this denominator.

A calendar-window rate answers another question. For a window W of positive length and a defined portfolio of tasks, let

$$V_W = \frac{\sum_{i:\, s_i + T_i \in W} S_i^0 R_i(\Delta)}{|W|}. \tag{7}$$

All counted acceptances need the same subsequent follow-up. Record work in progress at the window boundaries because some output may have begun earlier. Dividing portfolio output by the sum of overlapping task durations gives neither this calendar rate nor a team's capacity. Zero-effort tasks do not require individual output/effort ratios. A positive aggregate denominator and the count of such tasks are sufficient.

3.3. **The endpoints as functions of a marked history.** These endpoints can all be calculated from a task's history without specifying a probability law for execution. Write ω_i for its complete marked history, which records enrolment, attempts, review, first acceptance, abandonment, repair and follow-up assessments, together with the allocation functions in (3) and the frozen score. Each event mark identifies the event type and recorded outcome. The history need not be fully observed. The observation map $\mathcal{O}(\omega_i)$ represents telemetry by retaining selected events, interval totals or other records.

For fixed τ and Δ, the vector of path functionals is

$$\mathcal{R}_{\tau,\Delta}(\omega_i) = \big(H_i(\tau + \Delta),\ \min(T_i, \tau),\ \mathbf{1}\{T_i \leq \tau\},\ Q_i(\tau, \Delta)\big). \tag{8}$$

The components are labour in person-hours, restricted time without acceptance in hours, an acceptance indicator and retained reference output. Report them together while preserving

their separate meanings and units. In the last component, (5) grants credit for first acceptance at $T_i \le \tau$ and assesses the retained fraction at task age $T_i + \Delta$. Labour accumulates through the common task age $\tau + \Delta$. An accepted-state indicator cannot determine retention: a task may pass, lose functionality and be repaired before follow-up.

This representation permits dependence, changing attempt costs and history-dependent review. It introduces no Markov, stationarity or independence assumption. A realised labour total is point-identified by an observation record when it has the same value for every admissible complete history consistent with that record. This pathwise question differs from identifying a population parameter from an observed probability law, as in Proposition 2.1. The next section obtains the smallest and largest labour totals compatible with specified interval records. Appendices A and B add specific execution laws solely to illustrate divergence between the components of (8).

4. Attainable labour ranges from interval observations

An interval total leaves the timing of attention within it unresolved. We therefore seek all values of fixed-horizon labour compatible with the observations and the allocation constraint in (3). The *identified range* contains exactly those values: each is attained by at least one compatible history, and every compatible history gives a value in the range. This use of compatibility follows the general partial-identification perspective [15, Section 1]. We prove the allocation results directly. They describe what the observations determine, with sampling uncertainty and causal intervention effects requiring separate analyses.

4.1. **Bounding labour at a bin boundary.** Exact bin masses can bound the continuous labour target even when the distribution of work within each bin is unknown. Suppose a fixed cohort's labour is partitioned into complete, nonduplicated contributions on task-relative age intervals $[\ell_j, u_j)$, with $0 \le \ell_j < u_j$. Attribute shared work once. Each contribution arises from a nonnegative allocation integral with absolutely continuous cumulative labour and known total mass v_j in person-hours. Applying the bound requires complete coverage and nonduplicated attribution, which must be established independently of the interval totals. Ages and cutoffs use a common declared time unit, and masses must remain exact after conversion. Rounded weights require separate error bounds.

Proposition 4.1 (A continuous labour envelope)**.** *Under these premises, set $c = \tau + \Delta$ and $H(c) = \sum_i H_i(c)$. Then*

$$H_-(c) := \sum_j v_j \mathbf{1}\{u_j \le c\} \ \le H(c) \le \ H_+(c) := \sum_j v_j \mathbf{1}\{\ell_j < c\}. \tag{9}$$

The gap is $H_+(c) - H_-(c) = \sum_{j:\,\ell_j < c < u_j} v_j$. If $H_-(c) > 0$ and retained output $Q = \sum_i Q_i(\tau, \Delta)$ is known, then

$$\frac{Q}{H_+(c)} \le P_H(\tau, \Delta) \le \frac{Q}{H_-(c)}. \tag{10}$$

Proof. A bin ending by c contributes its full mass. A bin starting at or after c contributes zero. Absolute continuity excludes endpoint atoms. Every crossing bin contributes between zero and its mass. Summing gives (9) and the gap. Nonnegative Q and positive bounding denominators give (10) by division. □

When $H_- = 0$, the bounds need not establish a positive denominator or a finite upper productivity bound. If output instead has justified bounds $Q_- \le Q \le Q_+$, positive H_- gives $Q_-/H_+ \le P_H \le Q_+/H_-$. These bounds are conditional on exact bin masses. Sampling accuracy and nonresponse require separate assessment, and Horvitz–Thompson estimates need not equal complete-population bin masses. Causal confidence intervals require an additional comparison of interventions.

Example 4.2 (An interior cutoff)**.** One person's bin $[0, 2)$ contains one person-hour. At $c = 1$, work entirely in the first hour gives labour 1, whereas work entirely in the second gives 0. Both respect the allocation constraint. Uniform interpolation gives $1/2$, but the known bin mass alone supports the interval $[0, 1]$.

4.2. **Using the person's capacity.** The preceding envelope uses nonnegativity. Knowing which person supplied an interval mass also bounds how much can fit on either side of a cutoff. An aggregate across several people cannot use a one-person capacity without justification.

Proposition 4.3 (Attainable contribution from one interval)**.** *Let $\ell < u$, $c \in [\ell, u]$ and $v \in [0, u - \ell]$. Among measurable allocations $a : [\ell, u] \to [0, 1]$ with $\int_\ell^u a(t)\,dt = v$, the attainable values of $x = \int_\ell^c a(t)\,dt$ are exactly*

$$[\max\{0, v - (u - c)\},\ \min\{v, c - \ell\}]\,. \tag{11}$$

If $v < 0$ or $v > u - \ell$, no such allocation exists.

Proof. Nonnegativity and capacity on the two pieces imply

$$0 \le x \le c - \ell, \qquad 0 \le v - x \le u - c.$$

Their intersection is (11). Conversely, for any x in this interval, set $a = x/(c - \ell)$ before c and $a = (v - x)/(u - c)$ after c. Each value lies in $[0, 1]$ and gives the required integral. Omit a zero-length piece when c is an endpoint: the corresponding numerator is then zero. Finally, integrating $0 \le a \le 1$ gives the necessary condition $0 \le v \le u - \ell$. □

For example, $3/2$ person-hours in a two-hour interval imply between $1/2$ and 1 person-hours in its first hour. The coarser envelope is $[0, 3/2]$. Neither result justifies uniform interpolation. If $v = u - \ell$, capacity identifies every cutoff integral even though individual activity labels have not been supplied. For an interior subinterval $[b, c] \subseteq [\ell, u]$, the same construction on that subinterval and its complement gives the attainable marginal range $[\max\{0, v - ((u - \ell) - (c - b))\}, \min\{v, c - b\}]$.

Summing marginal ranges gives valid bounds for a nonduplicated partition of contributions. Attaining both summed endpoints requires compatible constructions for the component endpoints. Disjoint intervals of one person, or intervals belonging to distinct people, can be combined when no other constraints link them. Overlapping observations or several tasks competing for the same person's time require a joint calculation. For example, two task masses of one person-hour each on $[0, 2]$ individually allow $[0, 1]$ before time 1. Their sum is exactly one person-hour before time 1: the shared capacity is saturated throughout $[0, 2]$. Independently adding the marginal intervals would give only $[0, 2]$.

4.3. **An exact finite representation of compatible histories.** Fix finite sets of tasks $\mathcal{I}$, people $\mathcal{J}$ and roles $\mathcal{R}$, a cohort $\mathcal{C} \subseteq \mathcal{I}$, and a bounded calendar-time domain $[b_0, b_K]$ with $b_0 < b_K$ containing every observation interval and target window. An admissible history consists of measurable $a_{ijr} \ge 0$ satisfying $\sum_{i,r} a_{ijr}(t) \le 1$ almost everywhere for every person j. The declared roster and attribution rules define the labour being bounded. Their completeness requires evidence beyond the mathematical result. A residual category outside $\mathcal{C}$ can represent work that consumes capacity without contributing to the cohort target.

For observations $\nu = 1, \dots, M$, specify an interval $[\ell_\nu, u_\nu]$ within this domain, a set $G_\nu \subseteq \mathcal{I} \times \mathcal{J} \times \mathcal{R}$ of attributed streams, and finite bounds $v_\nu^- \le v_\nu^+$. Impose

$$v_\nu^- \le \sum_{(i,j,r) \in G_\nu} \int_{\ell_\nu}^{u_\nu} a_{ijr}(t)\,dt \le v_\nu^+. \tag{12}$$

Exact interval masses use $v_\nu^- = v_\nu^+$. Deterministic rounding or ascertainment-error bounds may instead justify a nonzero width. A sampled Horvitz–Thompson estimate is not an exact population mass and does not, by itself, justify either inequality.

Form the sorted distinct endpoints $b_0 < \cdots < b_K$ from the domain boundaries, every observation endpoint and every target boundary. For the cohort denominator, target boundaries include each cohort task's enrolment s_i and its fixed cutoff $c_i = s_i + \tau + \Delta$. Translate task ages into calendar time before partitioning, so that simultaneous work by the same person shares one capacity constraint. Retain cells outside a target window whenever an observation spans them. Put $d_k = b_k - b_{k-1} > 0$ and let x_{ijrk} denote the mass of stream (i, j, r) in cell k.

Proposition 4.4 (Exact cell reduction and attainable labour range)**.** *Under the preceding model, define the finite-dimensional feasible set $\mathcal{P}$ by*

$$x_{ijrk} \geq 0, \qquad \sum_{i,r} x_{ijrk} \leq d_k \qquad \text{for every } j, k, \tag{13}$$

$$v_\nu^- \leq \sum_{(i,j,r)\in G_\nu} \sum_{k:\,[b_{k-1},b_k]\subseteq[\ell_\nu,u_\nu]} x_{ijrk} \leq v_\nu^+ \qquad \text{for every } \nu. \tag{14}$$

For any prescribed linear target $h(x) = \sum_{i,j,r,k} w_{ijrk} x_{ijrk}$, the attainable target values over admissible histories equal $h(\mathcal{P})$. In particular, the fixed-horizon cohort denominator uses $w_{ijrk} = 1$ exactly when $i \in \mathcal{C}$ and $[b_{k-1}, b_k] \subseteq [s_i, c_i]$, and zero otherwise. If $\mathcal{P}$ is nonempty, its target range is the closed interval

$$[H_{\min}, H_{\max}] = \left[\min_{x\in\mathcal{P}} h(x),\ \max_{x\in\mathcal{P}} h(x)\right]. \tag{15}$$

Both endpoints and every value between them are attained. If $\mathcal{P}$ is empty, the observations and capacity assumptions are inconsistent.

Proof. Integrating any admissible history over each cell gives nonnegative masses. Integrating the shared capacity gives (13). Splitting each observation at the cell boundaries gives (14). The target is preserved because its weights are constant on cells.

Conversely, for $x \in \mathcal{P}$ set $a_{ijr}(t) = x_{ijrk}/d_k$ in the interior of cell k, and set the finitely many endpoint values to zero. These measurable allocations are nonnegative and have sum at most one for each person. They reproduce every cell mass, observation and target value. Thus the histories and the polytope have exactly the same target image.

The constraints define a closed convex set. Each coordinate is between zero and d_k, so a nonempty $\mathcal{P}$ is compact and a linear functional attains its extrema. Convex combinations of minimising and maximising points attain every intermediate value. The reconstruction just given supplies an admissible history for each of them. □

This is a pair of linear programmes with $|\mathcal{I}||\mathcal{J}||\mathcal{R}|K$ variables, $|\mathcal{J}|K$ shared-capacity constraints and $2M$ observation inequalities, in addition to nonnegativity. The construction preserves all attainable target values in the stated interval-integral model. It does so without approximating the allocation within a cell. Known cellwise restrictions, such as a person's unavailability, can be included if the same restrictions are imposed on reconstructed histories. Minimum session lengths, nonpreemption, smoothness and switching costs are different assumptions and are not established by this reduction. General bounded-error recovery studies similarly distinguish an admissible model and observations from a recovery rule [9, Sections 1.1–1.2]. The proof here is independent of the optimal-recovery theorems in that setting.

For exact retained output $Q \geq 0$ and $H_{\min} > 0$, monotonicity and attainability give the identified productivity interval $[Q/H_{\max}, Q/H_{\min}]$. Justified quality bounds $0 \leq Q_- \leq Q \leq Q_+$ give an outer interval $[Q_-/H_{\max}, Q_+/H_{\min}]$ containing every feasible ratio. It is the exact range only when every ratio in it is attainable from an admissible output–labour pair. Admissibility of the whole product $[Q_-, Q_+] \times [H_{\min}, H_{\max}]$ is sufficient. A zero lower labour bound does not justify dividing by it. The acceptance clock and retention assessment in (5) remain fixed throughout the allocation optimisation.

4.4. **One stream reduces to shortest paths.** For one scalar allocation $0 \le a \le 1$, cumulative masses give a smaller representation. At the sorted endpoints define $z_k = \int_{b_0}^{b_k} a(t)\,dt$. Write an inequality $z_q - z_p \le w$ as a directed edge $p \to q$ of weight w. Adjacent endpoints receive edges $k-1 \to k$ of weight d_k and $k \to k-1$ of weight zero. Observation bounds on $[b_p, b_q]$ add edges $p \to q$ of weight v^+ and $q \to p$ of weight $-v^-$. An exact mass v therefore gives edge weights v and $-v$. This is the standard difference-constraints representation [12, Slides 6–11].

Proposition 4.5 (Attainable cumulative bounds from graph distances)**.** *The scalar observations admit an allocation if and only if this graph has no directed cycle of negative total weight. When it is feasible, let $D(p,q)$ be the shortest-path weight from p to q. For any endpoint pair $p \le q$, the identified range of $\int_{b_p}^{b_q} a(t)\,dt$ is*

$$[-D(q,p),\ D(p,q)]. \tag{16}$$

Proof. Summing edge inequalities around a cycle gives 0 at the left, so a negative cycle is inconsistent. Conversely, adjacent edges make every vertex reachable from every other. In the absence of negative cycles, all shortest-path weights are finite and $D(r,r) = 0$. For any root r, potentials $z_k = D(r,k)$ satisfy each edge inequality because $D(r,v) \le D(r,u) + w$ for an edge $u \to v$ of weight w. Subtracting z_0 normalises the initial cumulative value without changing differences. Adjacent-edge inequalities give increments in $[0, d_k]$. Constant allocations with these increments satisfy the observations.

Summing inequalities along paths yields $z_q - z_p \le D(p,q)$ and $z_q - z_p \ge -D(q,p)$. Potentials rooted at p attain the upper bound, and those rooted at q attain the lower bound. Convex combinations of their reconstructed allocations attain the intervening values. □

The reference checker uses exact rational arithmetic and an all-pairs shortest-path calculation. Given indexed, sorted endpoints, graph construction and this calculation use $O((K+1)^3 + M)$ arithmetic operations and $O((K+1)^2 + M)$ storage. Bit costs depend on the encoded rational inputs. The checker compares the bounds and their reconstructed witnesses against independent finite allocation enumeration, including overlapping interval observations and inconsistent inputs. These finite constructed examples check the calculation against the proved result. Validation of a measurement process requires independent observations. The scalar graph does not enforce shared capacities between several task streams. That case requires the joint formulation in Proposition 4.4.

The application implements marginal interval formulas and sums eligible contributions, giving conservative aggregate bounds in exact arithmetic. With retained output treated as known and a positive lower labour bound, it also propagates these local labour bounds into a conditional cohort productivity range. Its reported `f64` endpoints are floating-point approximations, without a guarantee of outward rounding. The mathematical results and rational checker are exact under their premises. The joint linear programme gives the complete identified range for overlapping streams with shared capacity. The application does not implement that calculation.

5. Estimating labour from sampled allocations

Exact interval masses determine the feasible sets above. Probability samples support estimates whose properties follow from repeated application of a declared sampling design. Replacing an exact mass by a noisy estimate therefore changes the interpretation: the resulting deterministic bound need not cover true labour. We first establish the sampling estimator, then show how telemetry can assist it without assuming a correct prediction model.

5.1. **Estimating labour from observed allocations.** Activity sampling estimates labour from selected intervals without requiring continuous surveillance. Partition the observation

period into finitely many person-intervals indexed by m. Let d_m be interval length, $a_m \in [0,1]$ the target task's average allocation, and I_m indicate inclusion in a probability sample with known $\pi_m = \mathbb{E}[I_m] > 0$.

Proposition 5.1 (Design-unbiased labour accounting)**.** *Treat the allocations as fixed finite-population values. If sampled allocations are ascertained without error, then*

$$\widehat{H} = \sum_m \frac{I_m d_m a_m}{\pi_m}, \qquad \mathbb{E}_s[\widehat{H}] = \sum_m d_m a_m, \tag{17}$$

where $\mathbb{E}_s$ averages over the sampling design.

Proof. Linearity of expectation and $\mathbb{E}_s[I_m] = \pi_m$ give the equality term by term. Independence of inclusion indicators is unnecessary. □

This is the standard Horvitz–Thompson construction [11]. Its variance depends on the sampling design. Short diaries and task records can supply allocation estimates, while independent adjudication of a subsample can expose disagreement. Recording off-keyboard reasoning, review and coordination requires explicit categories. Inverse-probability weights do not remove nonresponse, inaccurate retrospective labels or work outside the sampling frame. Those errors need separate attention when telemetry is calibrated to sampled observations before and after intervention assignment. Missing AI-use records must remain missing rather than being coded as non-use.

5.2. **Frozen predictions as auxiliary information.** A probability sample can correct errors in telemetry predictions of total labour. The established difference estimator combines the predictions with a weighted sum of observed residuals. Denis and Haddou give the fixed-prediction form and its design variance explicitly [7, Section 2.3, equations (2)–(4)]. Prediction-powered inference also identifies its mean construction as a difference estimator and considers finite-population extensions [1, Section 1.4 and Appendix B of the accompanying preprint].

Let $U = \{1, \dots, M\}$ be a complete, fixed frame of person-intervals, with true labour $y_m = d_m a_m$ measured in person-hours and total $H = \sum_U y_m$. For every frame unit, retain a prediction f_m in person-hours. A raw SCH value requires a stated conversion before it can serve as f_m, so that prediction and observed labour have the same units. The entire prediction vector must be fixed with respect to the realised sample. Predictions trained on an independent earlier population can qualify, conditional on their frozen values and the subsequent sampling design. Accurate labels, a complete frame and correct inclusion probabilities remain substantive requirements.

Proposition 5.2 (Fixed-prediction difference estimator)**.** *For a probability sample $s \subseteq U$ with known first-order inclusion probabilities $\pi_m > 0$, set $r_m = y_m - f_m$ and*

$$\widehat{H}_{\text{diff}} = \sum_{m \in U} f_m + \sum_{m \in s} \frac{y_m - f_m}{\pi_m}. \tag{18}$$

If y_m and f_m are fixed under the design and every selected label is observed accurately, then $\mathbb{E}_s[\widehat{H}_{\text{diff}}] = H$. This holds without independence between inclusion indicators or correct prediction of individual outcomes.

Proof. Write $I_m = \mathbf{1}\{m \in s\}$. Linearity and $\mathbb{E}_s[I_m] = \pi_m$ give

$$\mathbb{E}_s[\widehat{H}_{\text{diff}}] = \sum_U f_m + \sum_U \frac{\mathbb{E}_s[I_m]}{\pi_m}(y_m - f_m) = \sum_U y_m.$$

□

The residual correction makes the estimate design-unbiased even when the prediction total is biased. Taking $f_m = 0$ recovers the Horvitz–Thompson (HT) estimator. Selection must include observed zeros as well as positive labour. A selected unanswered prompt is nonresponse, which must remain distinguishable from zero labour and from an unselected unit. The design-unbiasedness guarantee requires a complete frame, accurate allocations and known sampling probabilities. A provenance declaration alone cannot establish these properties. Independent label error with zero conditional mean may permit unbiasedness under both sampling and measurement error, but that requires an additional assumption beyond the fixed-label result proved here.

5.3. **Precision depends on residuals and the sampling design.** Let $\pi_{mk} = \mathbb{E}_s[I_m I_k]$, including $\pi_{mm} = \pi_m$. The exact design variance is

$$\operatorname{Var}_s(\widehat{H}_{\text{diff}}) = \sum_{m,k \in U} \frac{\pi_{mk} - \pi_m \pi_k}{\pi_m \pi_k} r_m r_k. \tag{19}$$

This follows by expanding the variance of $\sum_U I_m r_m / \pi_m$. The covariances require joint inclusion probabilities. First-order probabilities alone cannot determine them. Equation (19) is a population identity whose unsampled residuals are generally unknown. Estimating its value requires the actual sampling design and the joint-inclusion conditions needed by the variance estimator.

For simple random sampling without replacement of exactly n units from $M \geq 2$, every size-n subset has the same probability and $\pi_m = n/M$. Write

$$S_r^2 = \frac{1}{M-1} \sum_U (r_m - \bar{r}_U)^2, \qquad S_y^2 = \frac{1}{M-1} \sum_U (y_m - \bar{y}_U)^2.$$

The inclusion covariances in (19) give

$$\operatorname{Var}_s(\widehat{H}_{\text{diff}}) = \frac{M^2}{n} \left(1 - \frac{n}{M}\right) S_r^2, \qquad \operatorname{Var}_s(\widehat{H}_{\text{HT}}) = \frac{M^2}{n} \left(1 - \frac{n}{M}\right) S_y^2. \tag{20}$$

For $2 \leq n \leq M$, the unbiased sample-variance estimator is

$$\widehat{V}_{\text{diff}} = \frac{M^2}{n} \left(1 - \frac{n}{M}\right) s_r^2, \qquad s_r^2 = \frac{1}{n-1} \sum_{m \in s} (r_m - \bar{r}_s)^2. \tag{21}$$

Indeed, $s_r^2 = [n(n-1)]^{-1} \sum_{m<k,\, m,k \in s} (r_m - r_k)^2$. Each pair has inclusion probability $n(n-1)/[M(M-1)]$, so $\mathbb{E}_s[s_r^2] = S_r^2$. A noncensus sample of size one does not supply this variance estimate. A complete census has zero sampling variance, including a single-unit population. Measurement error remains a separate question.

For $n < M$, the difference estimator improves on HT exactly when $S_r^2 < S_y^2$. The gain in sampling precision therefore depends on residual variation under the design. Consider $M = 4$, $n = 2$, and $y = (0, 2, 4, 6)$. The true total is 12, and HT variance is $80/3$. Frozen predictions $f = (1, 3, 3, 5)$ yield variance $16/3$, whereas $f = (6, 4, 2, 0)$ yield $320/3$. Enumerating all six samples reproduces these variances and unbiased totals. It also verifies that (21) averages to each corresponding variance. The second predictor increases variance fourfold. These are finite mathematical checks, not evidence that a code-derived predictor improves precision in a workplace.

5.4. **Failure cases and implementation checks.** Fitting predictions on the realised sample invalidates the fixed-vector proof. For $M = 2$, $n = 1$ and $y = (0, 2)$, suppose a fitted rule memorises the selected outcome and predicts zero for the other unit. Its observed residual is always zero. The difference total is nevertheless 0 or 2, each with probability $1/2$, giving expectation 1 rather than $H = 2$. Generic cross-fitting is not a substitute for checking the conditioning and inclusion probabilities under the actual finite-population design.

The estimate can also be negative even though true labour is nonnegative. With $y = (0, 0)$, $f = (10, 0)$ and one uniformly selected unit, (18) is -10 or 10 with equal probability. Clipping

at zero changes its expectation from 0 to 5. Report the signed estimate and investigate an implausible result rather than silently changing the estimator. An unbiased labour denominator likewise does not make retained output divided by that denominator an unbiased productivity ratio.

The observatory's `estimate-labour` command requires a declaration that the frame is complete, stable unit identities, frozen predictions with their provenance, and positive inclusion probabilities. Each unit is recorded as unselected, observed or selected with nonresponse. It rejects nonresponse, missing freeze or frame attestations, duplicate identities, invalid probabilities and nonfinite arithmetic. A supplied general design supports point estimates only. Declared simple random sampling additionally requires the recorded selected count and probabilities to agree with n and M. It supplies (21) for $n \geq 2$ or zero sampling variance for a census. These consistency checks cannot establish that the declared sampling procedure was followed.

The implementation preserves signed estimates, observed zeros and unavailable uncertainty. It reports variance and standard error without a default confidence interval. An unbiased variance estimator alone establishes neither a normal approximation nor finite-sample interval coverage. Exhaustive synthetic tests cover unequal-probability sampling, both useful and harmful predictions, the sample-fitting counterexample, census and singleton cases, and invalid or missing observations. They verify the finite examples, input checks and reported estimates. Frame completeness, label accuracy and empirical calibration require evidence from the measurement process.

6. A study that could test the measurement proposal

The first empirical question is whether the proposed measurements can be obtained accurately and with the required coverage. A pilot should assess allocation ascertainment and interval coverage, then compare the precision of design-based estimates with and without fixed telemetry predictions. Independently adjudicated interval totals also permit comparison of labour identification intervals at prespecified cutoffs, including the reduction in width obtained from joint capacity constraints. This comparison requires exact masses rather than sampling estimates. The measurement results should inform a later intervention-effect study, whose precision will depend on independent teams, task variation and measurement error.

6.1. **A bounded prospective protocol.**

One implementable starting protocol recruits 12 teams with substantially distinct review queues, matched into six pairs using a two-week run-in. Matching uses repository maturity, task classes and baseline workload, all recorded before assignment. Within each pair, randomly assign one team to the specified AI intervention and one to the comparator. Enrol new tasks over six weeks. These numbers specify proposed recruitment and scheduling. Statistical power remains to be assessed. If review or agent capacity is shared extensively, the assignment unit must include that shared pool. Otherwise the proposed team randomisation does not isolate the intervention.

Use a predeclared class of maintenance and small feature tasks with a common intake process. Lock a baseline backlog before assignment and report it separately from later arrivals. Register each baseline task's enrolment relative to assignment and the treatment of any pre-assignment work, without silently resetting its clock or excluding effort. For new arrivals, apply the same eligibility and scope rules before implementers select tasks or expose them to the tools. Record all eligible requests, including those postponed or abandoned. This distinguishes an effect on execution of a fixed workload from a change in which work is undertaken because of the intervention. Report expanding scope as an outcome and retain the difference between the enlarged task and the original requirement.

For the initial pilot, take $\tau = 30$ calendar days from enrolment and $\Delta = 30$ days after acceptance. Maintain the assigned interventions and observation through the final required follow-up, 60 days after the last enrolment. The values may be unsuitable for other work

TABLE 1. Proposed measurements and the ambiguities each resolves. Collection details are to be registered before the pilot.

Record	Minimum content	Purpose
Intake	Requirement, scope version, enrolment, eligibility, frozen score	Preserve the unit and task mix before assignment
Labour	Role, task allocation, interval, sampling probability, nonresponse	Estimate all-role person-hours without double counting
Lifecycle	Start, attempts, review, acceptance, deployment, abandonment	Separate waiting, execution and unsuccessful work
Exposure	Assigned intervention, tools, versions, actual use, resource charges	Distinguish access, use and nonhuman cost
Follow-up	Common rubric, blinded audit, defects, repair and assessment date	Distinguish initial acceptance from retained functionality

classes and must be changed before that cohort begins. Document model versions, permitted autonomous actions, prompt and review practices, outages and comparator-tool access. Avoid a short crossover design: learning, code changes and queue effects may persist after a switch.

The minimal data record has five linked components, shown in Table 1. Task identifiers and participant roles connect the records. Identifiers used for analysis should be pseudonymous, with access to detailed work traces limited to the research purpose. The protocol need not capture private thought or treat keyboard activity as continuous labour.

For labour, a workable run-in samples four short intervals per participant-day at random from the declared work frame and asks for role and task allocation in each. Retain the inclusion probabilities and response timestamps. Independently adjudicate a random subset using diaries and task records, without disclosing intervention assignment where feasible, to assess disagreement. Measure response burden and coverage of after-hours work as well. The run-in variance and burden data can guide changes to sampling intensity before the trial protocol is frozen. Report persistent missingness by assigned intervention, role and task class, since expanding telemetry cannot by itself recover unobserved reasoning time.

For quality, assess a probability sample of accepted requirements at acceptance and after Δ using the same tests and review rubric in both groups. Blind reviewers to intervention assignment where the artefact permits it and record failures of blinding. Duplicate assessment of a subset allows the study to report agreement, patterns of disagreement and changes made by adjudication. Known assessment probabilities are needed to extrapolate retention scores to the full cohort.

Assessment can also change subsequent work. Prespecify when findings are fed back and count audit-triggered repair within the applicable labour boundary. If assessments change work, define intervention effects under common assessment and feedback procedures. Extrapolation from the audited subset then requires a design and assumptions accounting for audit exposure, beyond known assessment probabilities alone. Preserve severity-specific defects and actual repair labour as separate outcomes, because a single retained fraction cannot encode every consequence of a security or reliability failure.

6.2. **Estimands that retain unsuccessful tasks.** The primary arm-level summaries are mean all-role cohort labour, acceptance by τ, restricted time without acceptance, mean retained output and their aggregate productivity ratio. For potential outcomes under assigned intervention $a \in \{0, 1\}$, define

$$\begin{aligned} \mu_H(a) &= \mathbb{E}[H_i^a(\tau + \Delta)], \quad F_A^a(u) = \mathbb{P}(T_i^a \le u), \\ \mu_Q(a) &= \mathbb{E}[Q_i^a(\tau, \Delta)], \qquad P_H^a = \mu_Q(a)/\mu_H(a), \end{aligned} \tag{22}$$

with a positive labour mean for the last ratio. For the locked baseline backlog, the expectation weights each eligible task equally. Functional size enters only through S_i^0 in Q_i. Differences between intervention means and between their productivity ratios define effects of assignment, including any non-use of the assigned tools. Report the endpoints together: a reduction in labour could arise from abandoning more requests and would not, by itself, establish a benefit.

Later arrivals require a separate estimand because the intervention can change which requests arrive. A team-level adoption estimand instead uses a common calendar window and the full eligible portfolio, including changes in intake and work in progress, as in (7). Conditioning on treatment-induced submissions would change that target. Every contrast therefore needs its population and weighting stated.

Proposition 6.1 (Restricted time without acceptance)**.** *For any nonnegative extended acceptance time T, including $T = \infty$, and finite $\tau > 0$,*

$$M_A(\tau) = \mathbb{E}[\min(T, \tau)] = \int_0^\tau [1 - F_A(u)]\, du, \qquad F_A(u) = \mathbb{P}(T \le u). \tag{23}$$

Proof. For each outcome, $\min(T, \tau) = \int_0^\tau \mathbf{1}\{T > u\}\, du$. Tonelli's theorem permits exchanging expectation and the nonnegative integral. □

Smaller M_A means less time spent without acceptance during the specified horizon. Always report $F_A(\tau)$ alongside it. With complete follow-up to τ, empirical averages estimate both directly. Permanent abandonment remains nonacceptance in this definition. It is not dropped as a missing completion time. If abandonment is modelled as a terminal competing event, estimate acceptance cumulative incidence. Treating abandonment as independent censoring instead changes the target.

Administrative censoring before τ requires additional assumptions. Unconditional independent censoring permits ordinary survival estimation when there is no terminal competing event and observation support extends through the target horizon. Conditional independence requires the corresponding support within relevant at-risk covariate strata, together with conditioning or weighting that accounts for the conditional observation probabilities before marginal standardisation. Support means a positive probability of remaining observed wherever unaccepted tasks contribute to the target, including its endpoint. If support ends earlier, shorten the target horizon or state explicit extrapolation assumptions or bounds. Informative loss to follow-up requires a sensitivity analysis or bounds. A task that has not yet reached its follow-up date is not evidence of zero repair or perfect retention. The planned full follow-up reduces these problems but does not guarantee that losses will be absent.

6.3. **Analysis by assigned intervention.** A causal comparison based on randomisation applies to the implemented assignment under its stated interference assumptions. Shared reviewers, reusable AI-generated components and changes to colleagues' queues create spillovers. Cluster at a boundary that contains the main shared resources, record cross-boundary dependencies and describe residual interference.

Analyse all eligible tasks by their assigned intervention. Unequal team backlog sizes require care: a difference of pooled arm task means need not be design-unbiased for the fixed task-weighted contrast. For a locked backlog of known size $n_0 > 0$, let $\mathcal{I}_g$ be the fixed task set of team g, A_g its assignment, and $\pi_g(a) = \mathbb{P}(A_g = a) = 1/2$. For an observed task outcome Y_i, use the assignment-weighted mean

$$\widehat{\mu}_Y(a) = \frac{1}{n_0} \sum_g \frac{\mathbf{1}\{A_g = a\}}{\pi_g(a)} \sum_{i \in \mathcal{I}_g} Y_i. \tag{24}$$

Under consistency and the stated absence of cross-team interference, its assignment expectation is the finite-backlog mean of Y_i^a: replace each observed summand on $A_g = a$ by its potential outcome, then use $\mathbb{E}[\mathbf{1}\{A_g = a\}/\pi_g(a)] = 1$. The same argument applies to the

difference between intervention means. Labour and quality subsampling additionally require their own design weights and measurement assumptions. The estimate $\widehat{\mu}_Q(a)/\widehat{\mu}_H(a)$ remains a ratio estimator, not an unbiased ratio. Report it only with a positive denominator.

Inference must respect the matched-pair assignment and retain both assignment clusters and numerator–denominator dependence. Thousands of commits do not create thousands of independent treatment assignments. With only six pairs, randomisation-based comparisons are coarse and intervals may be wide. Report the pilot's feasibility and measurement results even when an effect estimate is inconclusive. Its estimates of clustering, outcome dispersion, attrition and assessment error, together with the run-in evidence, should inform the design of a larger trial around a prespecified precision or power target. The illustrative recruitment count alone establishes no claim of adequate power.

6.4. **Validating measurements and making a decision.** Validation must distinguish true task labour from its sampled estimate. Compare frozen-index predictions with sampled labour by assigned intervention, role and task class on held-out teams or periods. High correlation of binned means does not establish calibration, and sampling error changes how prediction errors should be interpreted.

Treat each task's labour H in a finite validation population as fixed. Let f be its prediction, frozen before validation sampling, and suppose $\mathbb{E}_s[\widehat{H}] = H$ with finite design variance. Then

$$\mathbb{E}_s[f - \widehat{H}] = f - H,$$
$$\mathbb{E}_s[(f - \widehat{H})^2] = (f - H)^2 + \mathrm{Var}_s(\widehat{H}).$$

Expand the square: the cross term vanishes because $\mathbb{E}_s[\widehat{H} - H] = 0$. Dependence between tasks' samples does not alter these marginal identities, but affects aggregate uncertainty.

For a second prediction g frozen before the same validation sample, subtraction cancels the common sampling variance:

$$\mathbb{E}_s[(f - \widehat{H})^2 - (g - \widehat{H})^2] = (f - H)^2 - (g - H)^2.$$

This permits a design-unbiased comparison of the two true squared losses. The identity does not, by itself, estimate either squared-loss level, absolute error or uncertainty in the comparison.

For example, eight one-hour person-intervals have allocations $(1, 1, 1, 1, 0, 0, 0, 0)$. Sample one uniformly, with inclusion probability $1/8$. True labour is $H = 4$, whereas its HT estimate is 0 or 8, each with probability $1/2$. A perfect prediction $f = 4$ has apparent absolute error 4 against the sampled estimate. The interval $[3.5, 4.5]$ covers true labour surely but covers the sampled estimate with probability zero. Mean residual calibration remains design-unbiased. Absolute error and interval coverage against $\widehat{H}$ concern the noisy observable.

For true-labour accuracy, use a denser, independently adjudicated validation subset, documenting remaining reference error, or a measurement model whose assumptions identify the target. Comparing predictions convolved with a known sampling-error law directly assesses the noisy observable. Recovering latent absolute error or coverage requires an identifying measurement model. Retain inclusion probabilities and joint sampling-design information for uncertainty calculations. The accompanying observatory reports HT point estimates from interval records. Its separate complete-frame estimator supplies a design-variance estimate only for the declared simple-random-sampling case under the conditions in Section 5.3.

The validation should also test how task splitting, squashing and representation affect the score, while reporting reviewer disagreement and follow-up completeness. Vary plausible missing labour and disputed retention scores to see whether they reverse an intervention ranking. Such a reversal makes measurement uncertainty part of the intervention decision.

These measurement uncertainties affect the intervention decision. Before examining outcomes, define the practical labour or timing improvement sought and a domain-justified

quality margin. Report the intention-to-treat endpoint contrasts with their uncertainty. Failure to detect a quality reduction is not evidence that a noninferiority margin is met. The results can support labour saving while leaving delivery speed unresolved, or reveal a trade-off between endpoints. Report retained-output productivity with its numerator and denominator, since an increase in their ratio alone cannot establish a staffing, quality or delivery benefit.

6.5. **Further modelling and observational alternatives.** For later forecasting, start with direct accounting and a small task-class baseline. Use a more elaborate model when it improves held-out predictive performance, with features available at the declared prediction time. The prediction target also determines the required modelling choices. Exponentiating a predicted mean of log labour does not directly recover arithmetic mean labour, because retransformation depends on the residual distribution. Tasks requiring zero human effort need an explicit treatment.

Staffing decisions may require a joint prediction of effort and time. Their marginal predictions alone leave the probability of meeting both a budget and a deadline unresolved without their dependence. A different joint quantity concerns the same task under both interventions. Even a randomised comparison of arm means does not identify the cross-intervention probability that it would improve under both unobserved counterfactuals.

If randomisation is unavailable, a staggered adoption design may use the cohort-specific estimands of Callaway and Sant'Anna [4], subject to its comparison-group, no-anticipation, overlap and conditional parallel-trends assumptions. A step in SCH must not itself define adoption timing. Reversible or changing use is not automatically a persistent adoption treatment.

7. Scope and next observations

The observation scheme determines which labour totals telemetry can identify. Commit events admit observationally equivalent activity laws. Exact interval masses and per-person capacity yield attainable cutoff bounds. Partitioning at every observation endpoint and target cutoff preserves all constraints needed to compute the complete identified range by finite optimisation. Separate interval bounds remain valid under their observation premises, but can omit restrictions imposed by shared capacity. An implementation that adds those bounds therefore reports a conservative range unless joint attainability is established.

Sampling adds uncertainty about the labour represented by unobserved intervals. Known inclusion probabilities give unbiased totals under accurate allocation ascertainment. A fixed telemetry predictor improves precision only when its residuals reduce design variance. When diaries supply allocation labels, their accuracy and coverage of reasoning and coordination require separate evidence. The sampling frame must also cover the target work. Identification ranges remain conditional on the measurements and allocation constraints. Design unbiasedness remains conditional on the sampling and ascertainment premises. A causal benefit from AI adoption requires a further comparison of interventions.

The measurement pilot should first establish whether task identities can connect all-role allocations to independently assessed retained output through the required follow-up. It should report ascertainment disagreement, missingness and collection burden so that a later intervention trial can choose its sampling intensity and precision target. A fixed-task trial and an organisation-wide adoption study address different quantities when the intervention changes incoming requirements. More detailed execution models become useful when observations can test their assumptions and improve a stated prediction or decision. The present results specify which quantities the available observations determine and which remain bounded or unidentified.

Appendix A. Illustrative mechanisms for effort and elapsed time

The main identification and sampling results require no execution law. The following small models illustrate why their distinct endpoints are needed. They give exact conditional comparisons for parallel execution, repeated attempts and stopping, without fitting these mechanisms to an organisation.

A.1. **Dependencies and parallel execution.** Represent the stages of one execution by a finite directed acyclic graph. Nodes have nonnegative elapsed durations. Edges encode all required precedence constraints. Let C be its longest path duration. Let H and G denote total human person-hours and homogeneous-agent instance-hours required by these stages.

Proposition A.1 (Resource and path bounds)**.** *With available capacities $m > 0$ people and $k > 0$ agents, every feasible schedule of elapsed length L satisfies*

$$L \geq \max\{C, H/m, G/k\}, \tag{25}$$

omitting a resource term when that class is absent. With unlimited resources and immediate execution of each ready node, the earliest completion time is C.

Proof. Every dependency path executes in order. Integrating resource capacity over the schedule gives $H \leq mL$ and $G \leq kL$. With unlimited resources, the earliest finish of a node is its duration plus the maximum predecessor finish time, with zero for an empty predecessor set. Induction along a topological ordering shows that the final finish is the maximum path duration. □

For finite resources, the maximum in (25) is only a lower bound. It does not solve the scheduling problem or describe queue interactions with other tasks. An agent-hour is comparable across nodes only within the declared resource class.

Consider one human production branch of duration $c_h \geq 0$ running independently beside one agent branch of duration $g \geq 0$, followed by human verification of duration $v \geq 0$. One attempt then consumes

$$h = c_h + v, \qquad t = \max(c_h, g) + v. \tag{26}$$

The maximum is justified by independent parallel branches. If a human stage needs the agent's result, its position in the graph changes and so does t. Figure 1 makes the person-time and elapsed-time distinction visible in a slow-agent example.

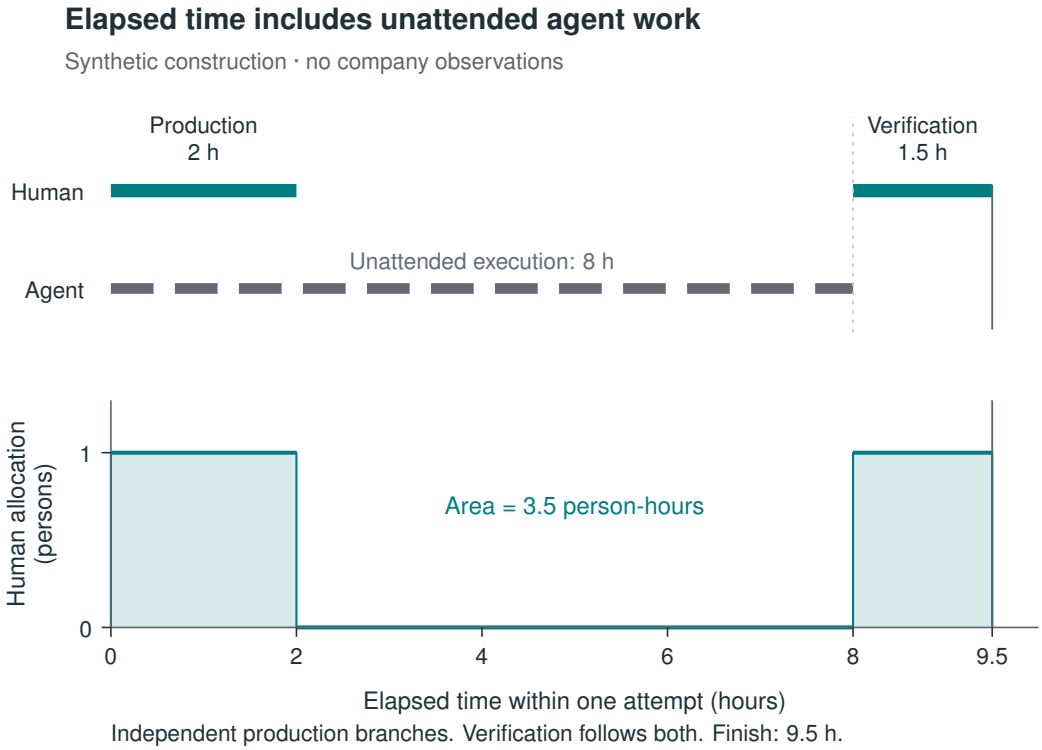


FIGURE 1. One synthetic attempt with human-production duration $c_h = 2$, agent duration $g = 8$ and verification duration $v = 1.5$ hours. Human production and agent execution begin together, and verification waits for both. Human labour is 3.5 person-hours and elapsed duration is 9.5 hours. The figure excludes the initial queue and subsequent retries.

A.2. **First acceptance under repeated attempts.** The cycle rates below describe eventual first acceptance. Comparing them with the study's fixed-horizon totals will show how an acceptance deadline changes the productivity ratio.

Assume independent, identically distributed pass/fail attempts with pass probability $p \in (0, 1]$, deterministic per-attempt labour $h > 0$ and duration $t > 0$. Attempts execute sequentially until the first pass, and N is their number. A one-off initial wait is $w \geq 0$. Write $H^A = hN$ for labour to first acceptance and $L = w + tN$ for elapsed time. These quantities exclude later repair.

Proposition A.2 (Retry moments and cycle rates)**.** *Under these assumptions,*

$$\mathbb{E}[N] = \frac{1}{p}, \quad \mathbb{E}[H^A] = \frac{h}{p}, \quad \mathbb{E}[L] = w + \frac{t}{p}, \quad \mathrm{Var}(H^A) = \frac{h^2(1-p)}{p^2}. \tag{27}$$

Suppose independent, identically distributed task cycles have fixed score $S^0 \geq 0$, bounded retained fraction R with $\mathbb{E}[R] = r$, and the same parameters above. Ratios of accumulated retained output to accumulated pre-acceptance labour and to serial cycle duration converge almost surely to

$$P_H^A = \frac{S^0 r p}{h}, \qquad V_{\text{serial}} = \frac{S^0 r}{w + t/p}. \tag{28}$$

Dependence of R on the attempt count within a cycle is allowed.

Proof. The tail probability is $\mathbb{P}(N > j) = (1-p)^j$ for integers $j \geq 0$, whose sum is $1/p$. For $p < 1$, conditioning on the first outcome gives $\mathbb{E}[N^2] = 1 + (1-p)(2\mathbb{E}[N] + \mathbb{E}[N^2])$,

so $\mathbb{E}[N^2] = (2-p)/p^2$. The moments are finite because the geometric tail decays. Hence $\mathrm{Var}(N) = (1-p)/p^2$. At $p = 1$, $N = 1$ surely. Scaling gives (27). For the rates, apply the strong law separately to the iid vectors (S^0R, H^A, L) and divide by the positive limiting mean denominators. This yields the ratios of expectations in (28). □

These elementary renewal-reward calculations [10] give ratios of accumulated totals. Expectations of task-specific ratios can differ. The serial rate applies to one repeated stream, so concurrent-team throughput requires a different model. The observation of R can occur after completion, provided all cycles are eventually assessed on the same basis. A wait repeated on every attempt belongs inside the multiplier $1/p$.

The eventual first-acceptance model also differs from (6) through its deadline: every modelled task eventually passes, whereas the cohort numerator credits only acceptance by τ.

Example A.3 (A horizon changes the ratio)**.** Set $p = 1/2$, $h = t = S^0 = R = 1$, $w = 0$ and $\tau = \Delta = 1$, with one person working continuously on sequential attempts until acceptance and no repair. Then $\mathbb{E}[Q] = 1/2$ and $\mathbb{E}[H(2)] = \mathbb{E}[\min(N, 2)] = 3/2$, so their ratio is $1/3$, while eventual-cycle productivity is $1/2$. These expectations follow from the probabilities of first-attempt success and second-attempt execution, each $1/2$.

A calendar deadline can also differ from an attempt cap when durations or waits vary. The first-acceptance labour boundary is narrower than (6). If $B_\Delta \geq 0$ is attributable labour during a specified post-acceptance follow-up with finite expectation, then the corresponding cycle productivity has denominator $h/p + \mathbb{E}[B_\Delta]$. That extension does not capture all labour in a fixed enrolment horizon automatically. Neither quantity includes unallocated shared overhead unless the measurement protocol adds it.

Corollary A.4 (Separate improvement conditions)**.** *For interventions $a \in \{0, 1\}$ satisfying the retry assumptions, expected labour and expected intrinsic duration decrease respectively if and only if*

$$\frac{h_1}{h_0} < \frac{p_1}{p_0}, \qquad \frac{t_1}{t_0} < \frac{p_1}{p_0}. \tag{29}$$

Here intrinsic duration excludes the initial wait. If $S^0 > 0$ and $r_0 > 0$, intervention 1 improves P_H^A if and only if

$$\frac{h_1}{h_0} < \frac{r_1 p_1}{r_0 p_0}. \tag{30}$$

Proof. Substitute (27) and (28), then multiply by the strictly positive denominators. The positivity of S^0 and r_0 is required for the last equivalence. □

Equal initial waits preserve the intrinsic-duration comparison. With unequal waits, compare $w_1 + t_1/p_1$ and $w_0 + t_0/p_0$ directly. A smaller drafting cost establishes none of these inequalities on its own.

Example A.5 (A retry reversal)**.** Changing (h, p) from $(1, 0.9)$ to $(0.5, 0.2)$ halves per-attempt labour but raises expected labour from $10/9$ to $5/2$ person-hours by (27). Lower effort per attempt can be overwhelmed by additional attempts.

A.3. **Stopping, heterogeneity and the limits of the mechanism.**

Proposition A.6 (A finite attempt cap)**.** *Under independent attempts with $p \in (0, 1]$, allow at most $K \geq 1$ attempts and count exhaustion without a pass as failure. The probability q_K of acceptance and expected attempt count m_K satisfy*

$$q_K = 1 - (1-p)^K, \qquad m_K = \mathbb{E}[\min(N, K)] = \sum_{j=0}^{K-1} (1-p)^j = \frac{q_K}{p}. \tag{31}$$

Let $r_K = \mathbb{E}[R \mid N \leq K]$ be the conditional mean retained fraction for acceptance within the cap. Expected labour is hm_K and expected retained output is $S^0 r_K q_K$.

Proof. Acceptance is the complement of K consecutive failures. The finite tail-sum identity gives the expectation, and multiplication of the geometric sum by $p = 1 - (1 - p)$ telescopes. The output formula follows by conditioning on acceptance. It assigns zero output to exhaustion while retaining its labour. □

At $p = 0$, the direct boundary definition gives $m_K = K$ and $q_K = 0$. At $K = 0$, both totals vanish and their ratio is undefined. Whenever $q_K = 0$, expected retained output is zero directly, so the conditional quantity r_K is not evaluated.

For $p > 0$ and $K \geq 1$, the ratio of expected retained output to labour is $S^0 r_K p/h$. Any remaining dependence of this ratio on the cap lies in r_K. Identically distributed accepted-quality marks independent of earlier attempts would give $r_K = r$ and a ratio independent of the positive cap. Without that extra condition, r_K may vary. Even when cancellation holds, the cap still changes the chance of delivering an individual request.

Example A.7 (Retention depends on the cap)**.** Let $p = 1/2$ and $R = \mathbf{1}\{N = 1\}$. Then $r = \mathbb{E}[R] = 1/2$ but $r_1 = 1$. At $K = 1$, true expected retained output is $S^0/2$. Substituting the uncapped r would incorrectly give $S^0/4$. Thus iid pass indicators alone do not make retained quality independent of the retry history.

Remark A.8 (Task heterogeneity). If per-attempt labour h is fixed and a task-specific pass probability P lies in $(0, 1]$, then, whenever $\mathbb{E}[1/P]$ is finite,

$$h\mathbb{E}[1/P] \geq h/\mathbb{E}[P]. \tag{32}$$

This follows from convexity of $x \mapsto 1/x$ and Jensen's inequality. Replacing heterogeneous probabilities by their mean can therefore understate effort. The conclusion does not extend unchanged to arbitrary dependence between effort and pass probability.

Learning from earlier failures can change both the cost and acceptance probability of later attempts. A retry model then needs history-dependent conditional probabilities, or direct totals that avoid that model. Shared queues, changing reviewer availability and abandonment also violate the elementary cycle description. Observations of these departures can determine whether the geometric model is suitable for the tasks being studied.

Appendix B. Illustrative detection effects and synthetic checks

An acceptance check is part of the observation process. More passing work can mean better production, easier assessment or both. To separate these mechanisms, suppose an attempt is defective with probability $d \in [0, 1)$ and the acceptance check detects a defect with sensitivity $s \in [0, 1]$. Correct attempts are never rejected. Attempts are independent with the same composition, and checking stops at the first pass.

Proposition B.1 (Detection changes the meaning of throughput)**.** *The pass probability and conditional correctness are*

$$p = 1 - ds, \qquad \rho = \mathbb{P}(\textit{correct} \mid \textit{pass}) = \frac{1 - d}{1 - ds}. \tag{33}$$

For fixed $S^0 > 0$ and $t > 0$, no initial wait and no repair cost, the serial recorded and correct-output rates are

$$V_{\text{recorded}} = \frac{S^0(1 - ds)}{t}, \qquad V_{\text{correct}} = \frac{S^0\rho(1 - ds)}{t} = \frac{S^0(1 - d)}{t}. \tag{34}$$

For $0 < d < 1$, lowering s strictly raises the recorded rate and strictly lowers correctness among accepted attempts.

Proof. Rejection requires both a defect and detection, so its probability is ds. Every correct attempt passes, and Bayes' rule gives (33). Summing over the number of earlier rejected attempts preserves the pass-conditioned composition at first acceptance. Equation (28) with

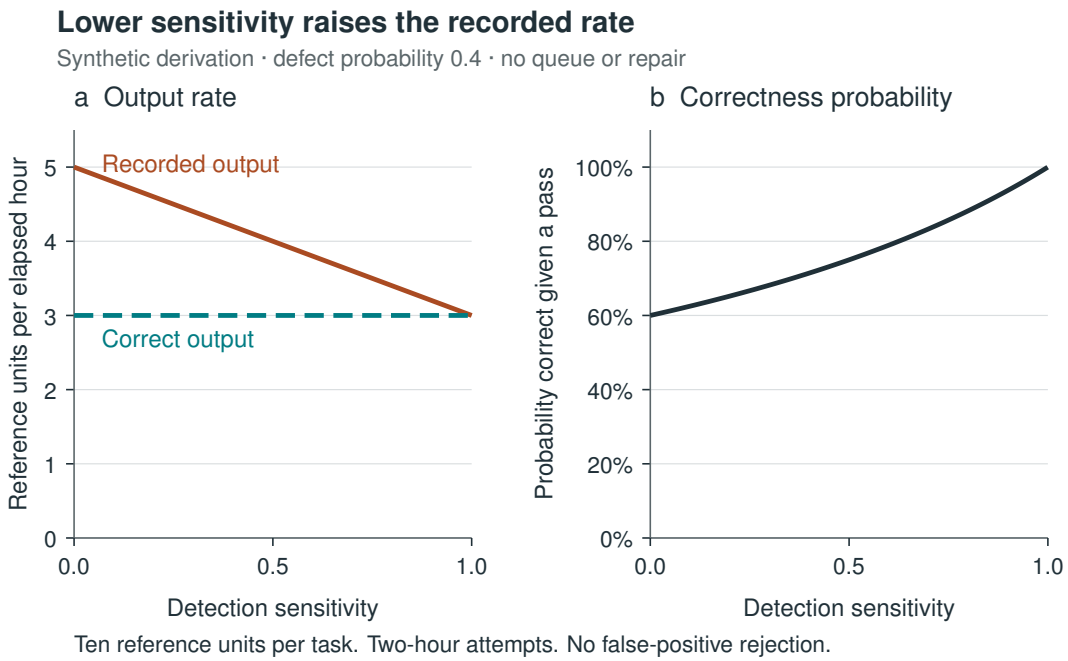


FIGURE 2. Consequences of the independent-attempt detection model with defect probability $d = 0.4$, attempt duration $t = 2$ hours and reference output $S^0 = 10$ units. Lower detection sensitivity raises recorded throughput and lowers correctness among accepted attempts. Correct-output throughput is constant only under the stated model, which omits escaped-defect consequences and repair. Conditional correctness ρ is distinct from the general follow-up fraction R.

$w = 0$, first with reward fraction 1 and then with ρ, gives (34). For positive d, lowering s increases the positive denominator of ρ while its numerator remains positive and fixed. □

The cancellation in correct-output rate depends on this mechanism's omissions. Escaped defects may impose repair costs, affect other tasks or violate an essential quality constraint, so the result gives no recommendation to weaken testing. At $d = 0$ there is no detection effect. False-positive rejection, heterogeneous attempts or nonconstant costs change the formula. To separate defect prevalence from detection sensitivity, the study needs an independent assessment sample: observed acceptance-check failures alone leave the two unresolved.

Table 2 and Figure 3 illustrate the independent effort and time conditions. Every task has $S^0 = 10$, one-off waiting time $w = 2$ hours and no post-acceptance repair. The four parameter tuples (c_h, g, v, p, r) are $(6, 0, 1, .95, .98)$, $(2, 1, 1.5, .90, .98)$, $(2, 8, 1.5, .90, .98)$ and $(1, .5, 7, .75, .90)$. These are chosen inputs, not estimates from developer data.

The shorter-production and slower-agent cases consume the same expected human labour, but only the former arrives sooner than the baseline. In the greater-verification case, reduced human production is outweighed by review and retries. The examples demonstrate logically possible mechanisms. They do not establish how often those mechanisms occur.

The executable supplement checks the rational arithmetic, counterexamples and finite-cap boundaries. Its Monte Carlo experiment uses seeds 17, 41 and 93 and 100,000 independent task draws per scenario per seed. It samples geometric attempt counts and human labour. Elapsed-time and retained-output values in Table 2 are analytic. From (27), for the mean

TABLE 2. Synthetic first-acceptance results from (26)–(28). Expected human labour $\mathbb{E}[H^A]$ is in person-hours, expected elapsed time $\mathbb{E}[L]$ is in hours including the initial wait, and productivity P_H^A is in retained reference units per person-hour. Productivity excludes repair.

Workflow	$\mathbb{E}[H^A]$	$\mathbb{E}[L]$	P_H^A
Human baseline	7.37	9.37	1.330
AI: shorter production	3.89	5.89	2.520
AI: slower agent branch	3.89	12.56	2.520
AI: greater verification	10.67	12.67	0.844

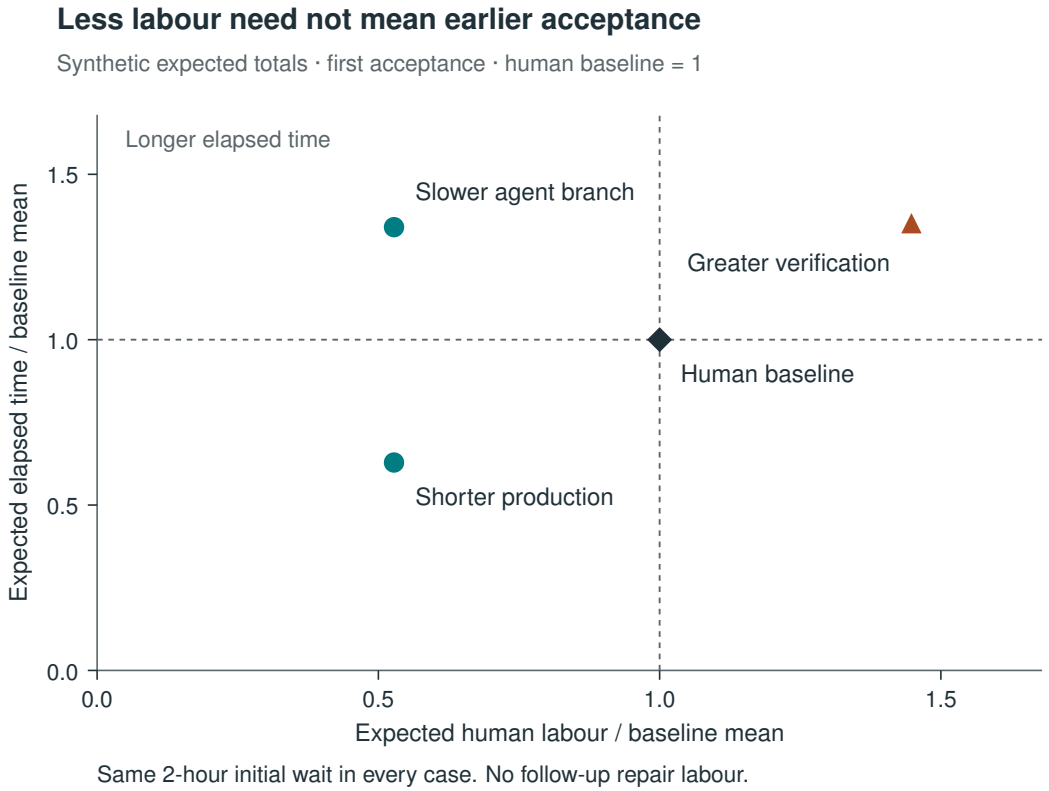


FIGURE 3. Synthetic expected human labour and elapsed time relative to the human baseline. The two equal-labour AI cases differ only in agent duration, demonstrating that lower human effort does not imply faster acceptance. The plotted values are exact model expectations rendered as decimals. They carry no empirical effect estimate or uncertainty interval.

$\overline{H}_n$ of n independent task-labour draws,

$$\sigma_n^2 = \operatorname{Var}(\overline{H}_n) = \frac{h^2(1-p)}{np^2}, \qquad \mathbb{P}(|\overline{H}_n - h/p| \le 10\sigma_n) \ge 0.99. \tag{35}$$

The probability bound follows from Chebyshev's inequality for $p < 1$ and exact equality of the mean for $p = 1$. It applies to each mean separately, without a simultaneous guarantee across seeds or scenarios. Agreement checks the implementation under its generating assumptions. Testing those assumptions, estimating a causal intervention effect and validating organisational measurements require empirical observations. Appendix E states which mathematical claims the formal proofs cover.

Appendix C. Targeted corrections to the Standard Coder argument

The following corrections concern identifiable statements in the seven-page version of [17]. They do not constitute reproduction of the fitted models or an exhaustive audit. Raw observations, model artefacts and split manifests were not supplied.

Internal agreement between two estimates of an activity state does not establish calibration to independently observed human labour.

Proposition C.1 (A consistency identity and its limit)**.** *For nested information sets $\mathcal{F} \subseteq \mathcal{G}$ and an integrable random variable Z, let $p = \mathbb{E}[Z \mid \mathcal{F}]$ and $q = \mathbb{E}[Z \mid \mathcal{G}]$ under the true joint law. Then*

$$\mathbb{E}[q - p \mid \mathcal{F}] = 0. \tag{36}$$

The identity does not imply that a pair of proposed estimates equals those conditional expectations.

Proof. The tower property gives $\mathbb{E}[q \mid \mathcal{F}] = p$. The reverse implication fails: a procedure returning the same constant $p = q = b$ has zero correction for any underlying activity process, including one whose conditional activity probability differs from b. □

Such a diagnostic can find incompatibilities within a model, but passing it does not establish activity calibration. Parameters used in a prospective prediction must also be fitted without future observations. Restricting the prediction inputs while fitting parameters to future events is still retrospective. The following arithmetic and inferential corrections concern the Standard Coder paper. Their validity does not require access to its unavailable empirical data.

Proposition C.2 (The stated sign-test rejection set)**.** *For $K \sim \text{Binomial}(10, 1/2)$,*

$$\mathbb{P}(K \in \{0, 1, 9, 10\}) = \frac{\binom{10}{0} + \binom{10}{1} + \binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{11}{512} = 0.021484375. \tag{37}$$

Proof. The four disjoint event masses have numerators $1, 10, 10, 1$, whose sum is 22. Divide by 1024. □

The stated probability in Section III.B of the source is 5.52%, whereas its stated rejection set gives approximately 2.15% under independent fair signs. A second binomial calculation for 24 positives among 300 tests gives an upper tail of approximately 5.42×10^{-8} with the corrected probability. This is conditional arithmetic, not an empirical rejection result: fair signs and independence within and across tests require justification. A mean-zero correction need not have equally probable signs. Neither calculation establishes calibration of hidden activity states.

Example C.3 (A correlation does not isolate efficiency variation)**.** Let X and Z be independent and uniform on $\{-1, 1\}$. Set $S = 2 + X$, $Y = 10 + 4X + 3Z$ and $U = 3X + 3Z$, so that $Y = 10 + X + U$. Then

$$\text{Corr}(S, Y)^2 = \frac{16}{25}, \qquad \frac{\text{Var}(U)}{\text{Var}(Y)} = \frac{18}{25} > 1 - \frac{16}{25}. \tag{38}$$

Proof. The independent centred signs have unit variance. Thus $\text{Var}(S) = 1$, $\text{Var}(Y) = 25$, $\text{Cov}(S, Y) = 4$ and $\text{Var}(U) = 18$, which give the displayed values. Both S and Y are strictly positive. □

Consequently, the residual fraction $1 - r^2$ cannot in general upper-bound the variance of a designated efficiency component. Such a bound needs a structural decomposition and appropriate orthogonality assumptions. Project averages do not themselves bound individual variation. The example supplies a logical counterexample, not a causal allocation of variance in data. Correlation also fails to establish calibration: multiplying an outcome by two leaves its positive correlation with itself equal to one.

Two reporting issues remain separate from these deductions. Figure 6's caption reverses the timing and code-feature provenance of target and prediction. Its approximate bin counts also imply about 1.2 million examples, compared with about 500,000 commits in the dataset description. An alternative counting unit or a typographical error could reconcile the latter. The unavailable manifest is needed to decide. Neither issue proves that the underlying observations or implementation are invalid. Feature substitutions in a fitted predictor are sensitivity calculations and need a separate identification argument before being interpreted causally.

Appendix D. Computing repeated cohort summaries

The accounting definition in (6) also determines a small computational problem. An analyst may compare many acceptance horizons on the same observed cohort while keeping the follow-up duration Δ fixed. Repeatedly scanning every record gives a simple reference implementation. Sorting the records once and storing cumulative totals replaces each subsequent scan with two binary searches. The optimisation preserves the reported quantities while changing their computational cost. Its runtime measurements do not establish an effect of AI assistance on human labour.

D.1. **Discrete input records and units.** The index receives two streams with different accounting roles. Let $A = ((a_i, w_i))_{i=1}^{n}$ contain one accepted-output record per accepted requirement. Each record gives an acceptance age a_i measured from that requirement's enrolment and a retained weight w_i assessed at the common follow-up Δ. Let $B = ((b_j, v_j))_{j=1}^{m}$ contain task-attributable labour contributions, including contributions from unfinished, rejected and abandoned tasks. Each b_j is the elapsed age from the associated task's enrolment to its contribution's integration boundary. Shared work is attributed before these records are formed.

Ages use a declared common integer time unit. All ages and weights are nonnegative `u64` integers, with output weights in millionths of a reference unit and labour weights in millionths of a person-hour. The two weight types are summed separately.

For nonnegative `u64` horizons τ and Δ in that time unit, the computational targets are

$$Q_{\mathrm{d}}(\tau) = \sum_{i=1}^{n} w_i \mathbf{1}\{a_i \le \tau\}, \qquad H_{\mathrm{d}}(\tau, \Delta) = \sum_{j=1}^{m} v_j \mathbf{1}\{b_j \le \tau + \Delta\}. \tag{39}$$

The result is the exact integer pair $(Q_{\mathrm{d}}, H_{\mathrm{d}})$. Their common scale cancels in the productivity ratio, which is defined only when $H_{\mathrm{d}} > 0$. A zero denominator remains an explicit zero in the pair. The Rust convenience method `ratio()` returns `None` in that case. With a positive denominator it returns a floating-point approximation for display. The integer pair remains the exact stored value.

For the discrete labour total to equal the continuous integral in (3), the selected contributions must sum to the labour accumulated by each queried cutoff. This equality holds when the contributions partition the cohort's labour into exact bin totals and no bin crosses a queried cutoff. Assigning a whole contribution to the end of a sampled interval or time bin instead approximates an interior cutoff.

The measurement protocol must also resolve missing labour, unobserved retention, duplicated tasks and scope changes, which the index cannot diagnose. Retained weights apply to the particular Δ at which they were assessed. A change to Δ, the cohort or its assessed weights requires corresponding input data and a rebuilt index. Section 4 states what exact interval masses and capacity restrictions establish at an interior cutoff. The discrete index below computes the supplied record totals. It does not solve the continuous optimisation for attainable labour bounds.

If the labour records contain exact, nonduplicated bin masses v_j on $[\ell_j, u_j)$ with end ages $b_j = u_j$, the Rust labour sum equals H_- in (9) after unit conversion. That index

Algorithm 1 Build exact cumulative totals for a fixed cohort and follow-up

Require: Accepted-output records A and labour records B as defined above, with ages and weights in the specified integer units.
Ensure: Sorted keys and exact prefix totals, or `TotalOverflow`.

for all stream names S in the ordered list (A, B) **do**
  $X_S \leftarrow \textsc{SortCopyByAge}(S)$
  $K_S \leftarrow$ age array of X_S; $P_S[0] \leftarrow 0$
  for $j \leftarrow 1$ **to** $|X_S|$ **do**
    $z \leftarrow \textsc{CheckedAdd128}(P_S[j-1], X_S[j].\text{weight})$
    if $z =$ overflow **then**
      return `TotalOverflow`
    end if
    $P_S[j] \leftarrow z$
  end for
end for
return $I = (K_A, P_A, K_B, P_B)$

Algorithm 2 Return exact output and labour totals for a batch of horizons

Require: Index I, horizons $\tau_1, \ldots, \tau_q$ and the fixed follow-up Δ.
Ensure: Exact total pairs in query order, or `HorizonOverflow`. No division is performed.

for $r \leftarrow 1$ **to** q **do**
  $c_r \leftarrow \textsc{CheckedAdd64}(\tau_r, \Delta)$
  if $c_r =$ overflow **then**
    return `HorizonOverflow`
  end if
end for
$O \leftarrow$ empty result array
for $r \leftarrow 1$ **to** q **do**
  $k_A \leftarrow \textsc{UpperBound}(K_A, \tau_r)$
  $k_B \leftarrow \textsc{UpperBound}(K_B, c_r)$
  Append $(P_A[k_A], P_B[k_B])$ to O
end for
return O

stores no start keys and does not compute H_+. A companion index could sort starts and ends separately, using inclusive end and strict start prefix lookups: $O(m \log(m+1))$ preprocessing, $O(m)$ storage and $O(1+\log(m+1))$ time per cutoff. The two totals then give the deterministic envelope under the stated observation premises. No runtime advantage for the companion calculation has been measured.

D.2. **Construction, queries and correctness.** The sorted prefix-index constructor, Algorithm 1, is a deterministic preprocessing algorithm for the two fixed record streams. The order of records with equal ages is immaterial: every such record is included or excluded together. Arrays use one-based record positions and a zero prefix. CheckedAdd128 widens the weight and either returns the exact sum or reports `TotalOverflow`. No wrapped or partial index is returned. The Rust implementation copies the input streams before sorting, so construction preserves the caller's records.

The batch horizon evaluator, Algorithm 2, is a read-only query algorithm on this index. $\textsc{UpperBound}(K, x)$ returns the number of sorted keys at most x, including every duplicate at the boundary. CheckedAdd64 reports `HorizonOverflow` if $\tau + \Delta$ cannot be represented. The entire batch is checked before any result is returned. Query order is preserved, so horizons need not arrive sorted.

Proposition D.1 (Agreement with a full scan)**.** *If index construction and horizon addition succeed, each returned pair in Algorithm 2 equals* (39)*. This includes empty streams, repeated ages, zero weights and zero labour totals.*

Proof. After construction has processed j records of a sorted stream, its loop invariant is $P_S[j] = \sum_{\ell=1}^{j} X_S[\ell]$.weight. It holds at $j = 0$ and checked addition preserves it. Sorting preserves the multiset of records. For a cutoff x, upper-bound binary search maintains a lower prefix whose keys are at most x and an upper suffix whose keys exceed x. Its remaining interval strictly decreases until its bounds coincide at k. Thus exactly positions $1, \ldots, k$ meet the inclusive cutoff. The corresponding prefix total is the same sum a full scan selects. Apply this argument at $x = \tau$ for output and $x = \tau + \Delta$ for labour. The empty prefix has total zero, so the boundary cases follow without division. □

This analytical proof establishes agreement for these discrete sums and operations. It is outside the Lean proof coverage in Appendix E. The Rust tests compare the indexed implementation with a separately implemented full scan. Successful tests do not establish completeness or reliability of the underlying observations.

With n acceptance records, m labour records and q queries, repeated scanning takes $O(q(1+n+m))$ time. Construction takes $O(n\log(n+1)+m\log(m+1))$ time and $O(n+m)$ additional storage. Indexed queries take $O(q(1+\log(n+1)+\log(m+1)))$ time and $O(q)$ result storage. These bounds count fixed-width operations and assume available memory. Preprocessing can outweigh the saving for a small batch, and an index cannot be reused after its input changes without maintaining or rebuilding its prefix totals.

D.3. **A reproducible performance comparison.** The supplement's dependency-free Rust program compares the scanning and indexed implementations on the same generated records and horizon batches. It retains the exact totals for equality checks and times computation separately from data generation. The benchmark includes construction time because a batch must save at least that time in queries to recover the cost of building the index. The fixed follow-up and synthetic inputs isolate the choice between scanning and indexing within the Rust program. They do not represent a company workload. For further reading on data science with Rust, see Van Der Post [16]. The runtime claims here rely on the accompanying measurements.

The workload contains 100,000 generated accepted-output records and 400,000 generated labour increments. It has no task roster or task-to-increment associations. Ages and horizons use hours, with $\Delta = 720$ and queried horizons from 0 to 1,440. Three fixed seeds (17, 41 and 93) define the datasets and horizon sequences. For each seed and each $q \in \{1, 10, 100, 1000\}$, five fresh processes each run one paired comparison. Each process first performs an unmeasured full scan followed by an unmeasured indexed run. Measured method order then alternates across observations. These are timings on warmed data and code, not cold-start measurements. Both implementations return identical integer pairs in every warmup and measured comparison. All 60 measured pairs are retained.

The build uses Rust 1.85.1, optimisation level 3, one code-generation unit, no link-time optimisation and enabled overflow checks, on a shared Linux host reporting an Intel Xeon Platinum 8573C processor. Dataset generation, startup, warmup, output verification and result I/O are outside both timed regions. Indexed total time includes input copies, sorting, prefix construction, queries and disposal of the index. The environment record retains the compiler details, source hashes and timing boundaries. Host load is uncontrolled. The 15 pairs per query count are repeated measurements of three synthetic datasets, not independent samples of production work.

Table 3 shows that indexing takes longer for a single query: median elapsed time increases from 2.889 to 21.016 ms. At ten queries, the median decreases by 26.0%, but individual paired reductions range from −16.4% to 55.1%. Indexing therefore takes longer in some of these comparisons. At 100 and 1,000 queries, every measured pair favours the index. For

TABLE 3. Measured cohort-query times in milliseconds: medians of 15 paired observations at each query count q. Indexed total is timed separately and includes index disposal. Component medians need not sum to it. Reduction is $100(1 - \text{indexed median/scan median})$, so a negative value means indexing takes longer.

q	Scan	Build	Query	Indexed total	Reduction
1	2.889	20.348	0.002	21.016	-627.4%
10	28.003	20.206	0.008	20.709	26.0%
100	263.415	22.348	0.082	23.046	91.3%
1000	2787.159	21.803	0.508	23.051	99.2%

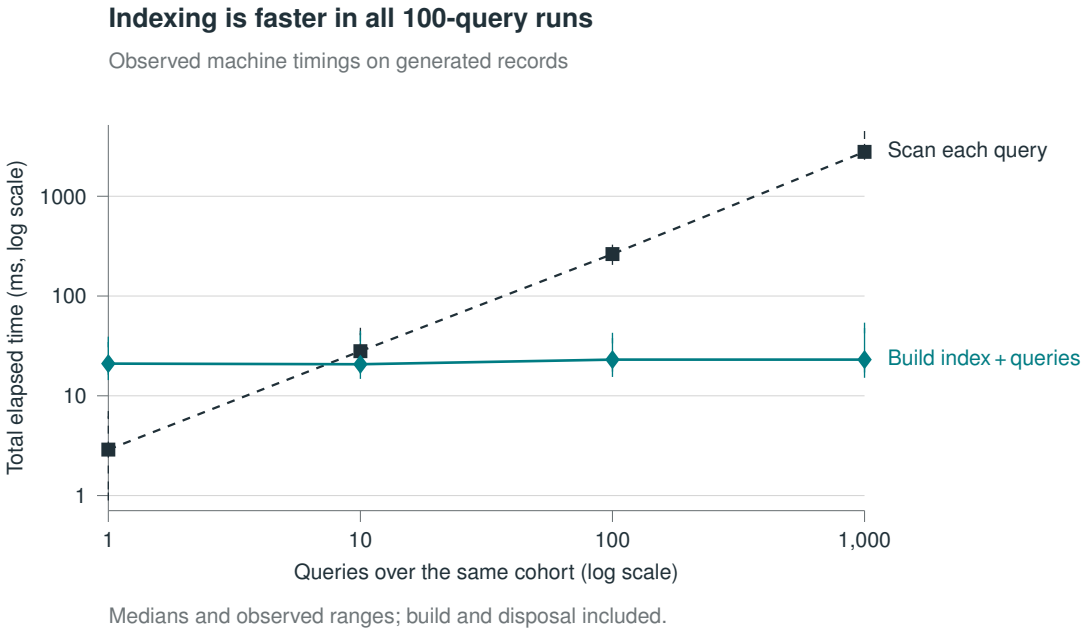


FIGURE 4. Observed elapsed times for repeated cohort-horizon queries on generated arrays of 100,000 accepted-output records and 400,000 labour increments. Markers are medians. Whiskers show the minimum and maximum across 15 paired runs (three seeds, five fresh-process repetitions per seed). They are not confidence intervals. Indexed totals include construction, queries and disposal. All one-query pairs favoured scanning and all 100-query pairs favoured indexing. Ten-query pairs varied in direction. Only four query counts were timed, so the connecting lines do not locate break-even. This result is confined to the generated workload and execution environment and does not estimate human labour or an effect of an AI intervention.

1,000 queries, the median falls from 2,787.159 to 23.051 ms, a 99.2% reduction including construction and disposal.

Figure 4 shows the observed ranges as well as the medians. The ranges describe variation across seeds and repeated host runs. They are not confidence intervals. The tested query counts do not identify an exact break-even point.

The observed saving is consistent with amortising preprocessing over repeated scans. It supports the index for this fixed workload and query pattern, subject to its storage and update costs. It does not establish performance on larger-than-memory datasets, continuously changing input records, different hardware or different query distributions. Raw paired

timings, the generator, exact-output checks and the summary script are retained so that another workload can test those boundaries.

Appendix E. Proof coverage and reproducibility

The Lean 4.33.1 project retained with the 13 October 2026 manuscript is pinned to mathlib v4.33.1 and contains 43 named theorems, including supporting lemmas. Its receipt records a successful source build, each declaration's assumptions and source hash, and an audit of every exported theorem's axioms. The declarations use only the recorded standard Lean axioms `propext`, `Classical.choice` and `Quot.sound`, where required. No admitted proof, custom axiom or native-decision shortcut is used. The original manuscript, proof sources and receipt remain together in the unchanged baseline package.

The marked-history representation, attainable labour bounds, model-assisted sampling results and sampling-validation argument are outside that receipt's coverage, as are Appendix D, its algorithms and its Rust implementation. These additions have analytical arguments and executable checks. The table below states what the retained Lean proofs establish and which related claims remain outside them.

Article claim	Checked content and remaining boundary
Proposition C.2	The actual ten-sign space has 1,024 elements. The rejection event has 22. Its uniform probability is 11/512. Fairness and independence of empirical signs remain assumptions.
Proposition 2.1	Explicit hidden-state marginalisation gives equality for every finite binary string at the two rational parameter pairs, and their finite-history expected activity differs for every positive length. The infinite-process construction and interpretation as labour remain analytical.
Example C.3	The actual four-point means, covariances, variances, positivity and decomposition are checked. The designated component has no established causal interpretation.
Proposition A.2	Convergent geometric weighted sums establish total mass, first and second moments, central second moment, scaled labour variance and waited duration. The iid retry interpretation and the strong-law limit of cycle ratios remain analytical.
Proposition A.6	A finite weighted tail identity, its geometric-tail specialisation and closed form are checked, including $p = 0$, $K = 0$ and $p = 1$ boundaries. Acceptance and retention assumptions are not inferred from the identity.
Corollary A.4 and Proposition B.1	Positive-denominator equivalences, a normalised nonnegative defect/detection table, conditional correctness, rate cancellation and strict detection trade-offs are checked. First-pass composition requires the stated probabilistic argument. Empirical adequacy requires independent observations.
Propositions 5.1 and A.1	Finite inverse-probability weighting and the capacity-bound algebra are checked. Measurement accuracy, the schedule-to-capacity premises, the DAG recursion and the causal interpretation of (24) remain separate.

The `Basic`, `Binomial` and `Geometric` modules contain the proofs. The complete inventory and semantic map in the supplement distinguish these statements from deferred conditional-expectation, strong-law, Jensen and restricted-time integral proofs. An analytical proof is not relabelled as machine-checked merely because a related arithmetic lemma compiles. The Python calculations are tested implementations, not formally proved refinements of the Lean definitions.

The original supplement retains the earlier generic numerical regression benchmark for provenance. It is not a validation of the task model, the empirical study or causal inference, and it supplies no performance result used in the article. Reproducibility requires the

source, fixed synthetic inputs, seed record, software environment and build logs. A successful computation or proof check establishes only the claim encoded by that artefact.

The 13 October 2026 baseline was developed through a communicating multi-agent workflow using MAS Math Lab 0.3.0, its existing Rust/nom structured-message codec, and separate drafting, mathematical, source-audit and checking contexts. The earlier delivered exchange was retained as an input. The protocol was not newly invented or adapted for this article. Agent reviews and Lean checks are disclosed development aids, not external peer review. The 2019 source paper is the work of its cited authors. The retained package preserves the correction records and execution limits. The complete development message archive is not included. Empirical validation and specialist review remain necessary before submission.

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